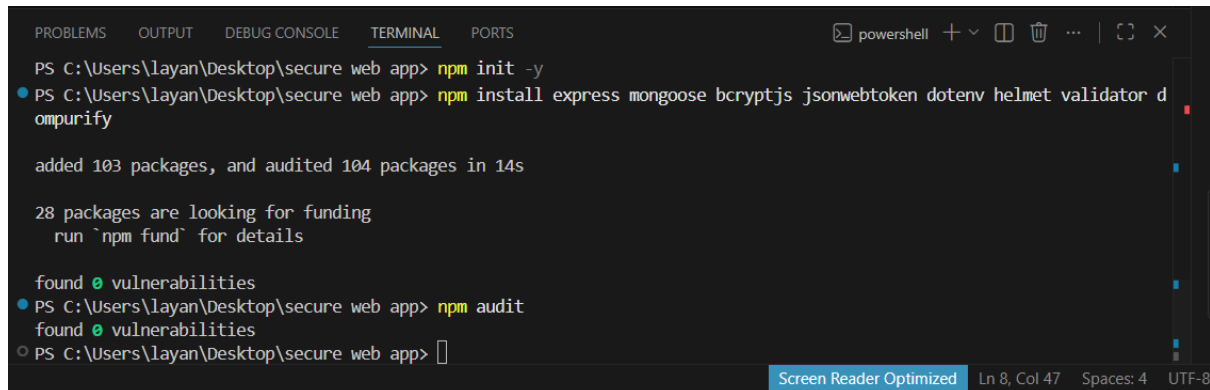




```
PS C:\Users\layan\Desktop\secure web app> npm init -y
PS C:\Users\layan\Desktop\secure web app> npm install express mongoose bcryptjs jsonwebtoken dotenv helmet validator dmpurify

added 103 packages, and audited 104 packages in 14s

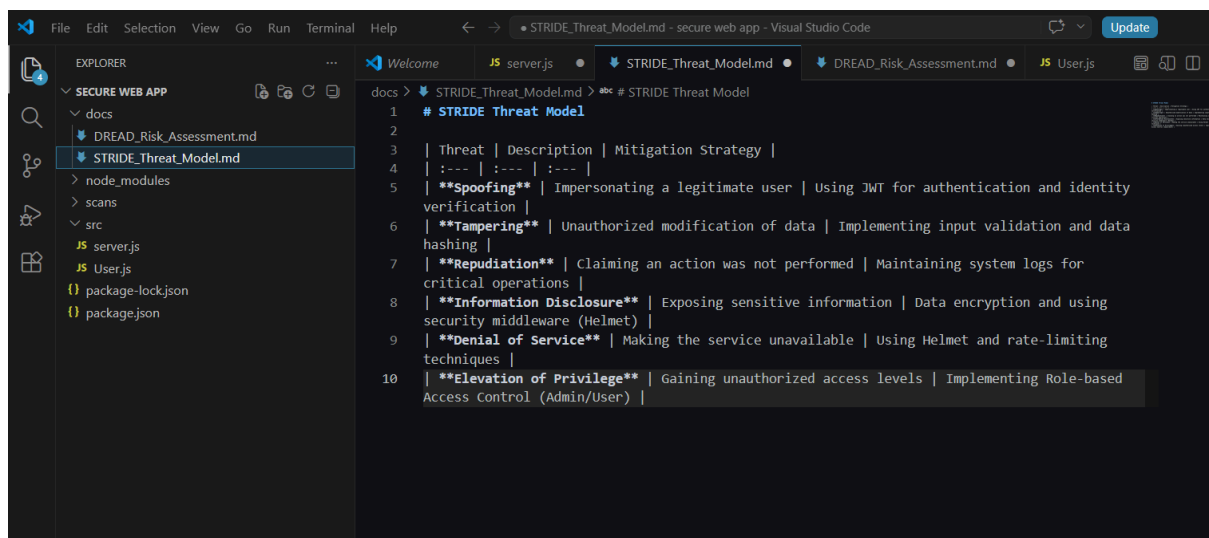
28 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\layan\Desktop\secure web app> npm audit
found 0 vulnerabilities
PS C:\Users\layan\Desktop\secure web app> 
```

Security Scan Report:

The above screenshot displays the output of the `npm audit` command performed on the project. The result "0 vulnerabilities" confirms that the installed dependencies are secure and free from known vulnerabilities.

STRIDE :



```
1 # STRIDE Threat Model
2
3 | Threat | Description | Mitigation Strategy |
4 | :--- | :--- | :--- |
5 | **Spoofing** | Impersonating a legitimate user | Using JWT for authentication and identity verification |
6 | **Tampering** | Unauthorized modification of data | Implementing input validation and data hashing |
7 | **Repudiation** | Claiming an action was not performed | Maintaining system logs for critical operations |
8 | **Information Disclosure** | Exposing sensitive information | Data encryption and using security middleware (Helmet) |
9 | **Denial of Service** | Making the service unavailable | Using Helmet and rate-limiting techniques |
10 | **Elevation of Privilege** | Gaining unauthorized access levels | Implementing Role-based Access Control (Admin/User) |
```

DREAD :

FileEditSelectionViewGoRunTerminalHelp

• DREAD_Risk_Assessment.md - secure web app - Visual Studio Code

Update

EXPLORER

SECURE WEB APP

docs

DREAD_Risk_Assessment.md

STRIDE_Threat_Model.md

node_modules

scans

src

server.js

User.js

package-lock.json

package.json

docs > DREAD_Risk_Assessment.md > # DREAD Risk Assessment

1 # DREAD Risk Assessment

2

3 | Threat | Damage | Reproducibility | Exploitability | Affected Users | Discoverability |

4 Total | Risk Level |

5 | :--- | :---: | :---: | :---: | :---: | :---: | :---: |

6 | SQL Injection | 8 | 7 | 8 | 9 | 8 | 40 | High |

7 | Session Hijack | 7 | 7 | 6 | 8 | 7 | 35 | High |

8 | XSS Attack | 6 | 5 | 5 | 7 | 6 | 29 | Medium |

9 | Brute Force | 4 | 3 | 2 | 4 | 3 | 16 | Low |